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Separated by a Common Language: How the Nature of Code Differences Shapes Communication Success and Code Convergence

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Abstract. Coordinated action within and between organizations is easier when individuals share *communication codes*—mappings between stimuli and labels. Because codes are specific to the groups within which they arise as conventions, collaboration across organizational units that have developed their own distinctive codes is often difficult. However, not all code differences are equal in their implications for communication difficulty and the capacity of individuals starting out with different codes to develop a shared code. Using computational models, we develop a theory about the nature of differences in initial communication codes and how they impact convergence on a common code. Our results show that the difficulty of code convergence lies not as much in learning new codes as in unlearning existing ones. The most severe challenges to communication stem from “code clashes” where codes contain different mappings between the same labels and stimuli. Furthermore, clashes that arise when agents have developed their individual codes in different task environments but draw on a common set of labels are likely to be the hardest to recover from, reflecting the perils of being “separated by a common language.”

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1. Introduction

Division of labor leads organizational units to develop specialized *communication codes* that map labels (e.g., words, images, descriptions) to stimuli (e.g., objects, events, perceptions) arising in their particular task environments (Arrow 1974, Brennan and Clark 1996, Clark 1998). However, integration of effort and output across unit boundaries often requires communication, which is likely to break down without a shared code (Monteverde 1995, Reagans et al. 2016). Communication breakdowns caused by divergence in codes recur across a range of organizational contexts, including intrafirm projects that cut across geographic or functional boundaries (Bechky 2003, 2006), recently merged firms (de Vecchi 2012, Graebner et al. 2017), and interfirm relationships (Tushman and Katz 1980, Zander and

Kogut 1995, Gulati and Singh 1998). Misunderstandings hinder collaboration, leading to delays, missed opportunities, errors, waste, and conflict (Korovyakovskaya and Chong 2015).

Although the occurrence of miscommunication because of differences in codes across organizational units is not surprising, the persistence of these breakdowns remains puzzling. Computational models (Skyrms 2010, Spike et al. 2017) and experiments (Brennan and Clark 1996, Weber and Camerer 2003, Selten and Warglien 2007) show code differences to be surmountable through repeated interactions that reward coordination. However, studies in the field document examples of significant code differences even after extensive interaction (Snook 2002, Jones 2003, Estival and Molesworth 2011). Misaligned interests do not appear to be the

culprit, as many of the well-documented instances of persistent communication breakdowns are from field settings in which interests are strongly aligned toward successful communication, such as civil aviation (Jones 2003, Howard 2008), military operations (Snook 2002), and healthcare (Ganiyat et al. 2013). Moreover, in many cases of persistent communication breakdowns, all parties speak the same natural language (Snook 2002, Cr  mer et al. 2007, Edwards et al. 2011). The surprising difficulties of communication in such situations bring to mind the witticism (attributed to George Bernard Shaw) that “the United States and Great Britain are two countries separated by a common language” but remain unexplained.

In this paper, we develop a theory that aims to explain *when we should expect communication breakdowns to be more (or less) persistent across organizational boundaries and what we can do about them*. We build on the insight that, although communication codes in use within a group are arbitrary in associating labels to stimuli (Arrow 1974, de Saussure 1998), differences that arise between codes are not arbitrary, neither in their origins nor in their consequences. There is systematic variation in the kinds of differences that arise between codes when members of different groups interact, which lead in turn to predictable variation in persistence of communication failures, even if there are no incentive misalignments and all parties are motivated to communicate successfully and are willing to learn how to do so.

Specifically, we propose that code differences fall into two broad types: codes may contain associations between different sets of labels and stimuli, or they might have the same set of labels and stimuli but map them differently. The first type of difference arises between codes that have developed in mutually unfamiliar contexts. For instance, communication within organizations often spans functionally specialized departments (e.g., information technology (IT) support to business units), making it likely that stimuli familiar in one code may be absent in the other. Functional departments are also social groups that are likely to use different vocabularies, making the labels in their codes different as well. The phrase “zombie debt” is relevant for the debt collection department of a credit card business but may mean nothing to an IT worker for whom neither the phrase nor the concept of statute of limitations are familiar. In such cases, both stimuli and labels in one code will be *alien* to the users of the other code.

The second type of difference arises across social groups within the same broad linguistic community and work environment. Speakers of the different codes use the same labels and are familiar with the same set of stimuli but map these differently; the codes are *jumbled*. For instance, the same business terms may be associated with different meanings in different organizational units even if they are not functionally

specialized. Individuals in one regional branch of a bank may use the term “attrition” for departure of employees, whereas those in another use it to describe the loss of customers (relying instead on “turnover” for employee departure, which in turn may be used to refer to sales in the first unit). Although the same set of stimuli and labels are used in the different codes, the jumbled associations can give rise to miscommunication. Indeed, Gulati et al. (2008) observe such differences in codes to seriously impede data aggregation efforts within corporations.

Our analysis, using computational models, demonstrates that differing associations among familiar stimuli and labels (jumbled codes) create a more persistent barrier to code convergence than unfamiliar stimuli or labels (alien codes). The key mechanism is the differential challenge of unlearning across the types of initial codes. Code differences unerringly lead to communication failures, but these can be overcome when existing associations in codes are unlearned and new associations are formed. Learning alien codes with unfamiliar stimuli and labels do not require unlearning; new associations can be acquired without much interference from prior knowledge. Convergence of jumbled codes, however, requires unlearning of prior associations that clash. In other words, it is clashing associations rather than mere differences in codes that make unlearning, and therefore eventual convergence, difficult.

Furthermore, we find that all clashes in jumbled codes do not pose equivalent difficulties. Codes can clash when they associate different labels to the same stimuli (label clash), different stimuli to the same label (stimulus clash), or both. Label clashes are likely to arise in interaction between agents that belong to different linguistic communities but share a task environment. For instance, de Vecchi (2012), studying the merger of two French airlines, finds that crews used different terms to refer to a common object, such as a food cart (“trolley” versus “comodi”). In academia, Suddaby (2010, p. 352) complains about the “proliferation of different terms and labels for similar phenomena—a problem that is often colloquially described as putting ‘old wine in new bottles’.” In contrast, stimulus clashes arise in interactions between agents that develop technical jargons for their specific tasks while drawing on the same vocabulary (Clark 1998). For instance, Bechky (2003) documents a communication breakdown between engineers and assemblers who had different associations for the word “slide.” Stimulus clashes also appear with some regularity in academia, giving rise to “science friction” when scholars in different disciplines attempt to collaborate (Cr  mer et al. 2007, Edwards et al. 2011).

We find that, although both label and stimuli clashes create communication challenges, on-the-job

interaction, using labels to communicate about stimuli as they arise in the environment, can relatively easily correct the former but not the latter. Although prior research has treated use of the same labels as indication of cultural cohesion (van Swol and Kane 2019) and has used label clashes to create miscoordination (Reagans et al. 2016), it is in fact codes with stimulus clashes that create a greater barrier to code convergence. For instance, both “comodi” and “slide” create miscommunication when individuals with the different codes interact. However, although a person unfamiliar with “comodi” only learns to associate it with a food cart and need not unlearn their prior label trolley the use of slide can continue to cause miscommunication for longer as both parties unlearn their associations.

Our results offer one explanation for the perils of being separated by a common language: clashes between codes that share stimuli or labels (and especially those that share labels but not stimuli) are hardest to resolve for any given level of code difference. This can potentially make cross-functional collaboration within a firm more challenging than cross- or within-functional collaboration between firms. We conclude that when stimulus clashes are suspected, that is, when communication occurs between groups that operate in different task environments but create labels from the same vocabulary, organizations may be better off using interfaces (such as translators who are aware of code differences and are able to switch between multiple codes) or alter workflows to reduce the need for communication.

This study supplements the view that firm boundaries are points of discontinuity in communication difficulty (Arrow 1974, Monteverde 1995, Kogut and Zander 1996) by revealing heterogeneity in code differences and resulting failures in communication, some of which may exist even within firms. Our typology of code differences allows us to predict when communication difficulties are likely to be the most severe by drawing on knowledge of the settings in which codes emerged. This can help organizational actors anticipate the severity of communication challenges across organizational boundaries (within and across firms) and take appropriate measures.

In relation to research on the integration of culture (Shrivastava 1986, Weber and Camerer 2003, Schein 2006, Chatman and O'Reilly 2016) and knowledge (Cronin and Weingart 2007, 2019; Garicano and Wu 2012), our study shows that unlearning is a necessary component of integration when codes clash, which is likely when codes develop in similar, rather than alien, contexts. It is not nonsharedness among initial codes that is a problem, but differences in mapping among the shared elements that cause clashes, miscommunication,

and potential for conflict. Individuals from entirely different cultures, alien to each other, may paradoxically have some advantages at eventual convergence over people with similar backgrounds (assuming a shared interest in communicating and the same cues and feedback in all cases).

Our models also offer a foundation for further computational studies on the interaction and coevolution of language and identity. Prior studies show that membership in social identities can be marked and made salient by use of particular communication codes (Bourhis and Giles 1977), creating opportunities for signaling, enacting, or constructing identities through the use of particular codes (Bucholtz and Hall 2004, 2005). At the same time, motivations to communicate within or across groups defined by particular social identities can shape how codes evolve (Auer 1999). Our study suggests that the nature of code differences, stemming from the similarities and differences of contextual origins of codes across identity groups, can impact both processes. We conclude the paper with a few sketches of such extensions and questions for future research.

2. Theory

2.1. Communication Codes in Organizations

Coordination in organizations often relies on communication, which can be seen from a semiotic perspective as having three levels: (1) syntactic, transmission and recognition of a message; (2) semantic, comprehension of the intended meaning of the message; and (3) pragmatic, interpretation of the message in terms of relevance and agreement (Shannon and Weaver 1949, Chandler 2017). Difficulties at all three levels create challenges for communication in organizations (Daft and Lengel 1986). At the syntactic level, information may resist encoding (von Hippel 1994). At the pragmatic level, individuals embedded in particular practices may resist others' point of view (Carlile 2002, 2004). However, organizational communication can break down even with no syntactic and pragmatic challenges in a world of perfect transmission and agents that are inclined to cooperate. This is because semantic alignment cannot be assumed, even within the same organization.

Communication success at the semantic level depends on there being sufficient overlap between the code that one individual uses to encode a message (attaching labels to stimuli or concepts) and the code that another individual uses to decode it (identifying the intended stimulus and its mental representation).¹ Because codes are arbitrary and conventional (rather than natural), the similarity of individuals' codes depends on them learning the same code (de Saussure 1998). The same labels may generate different

meanings for different individuals, as everyone learns the codes that are used within their socio-cultural contexts and codes that are seen as appropriate for their positions and roles within those contexts.² The conventionality of codes and their specificity to social contexts also implies that groups develop codes that suit their communication needs. Codes vary because group needs vary. As de Saussure put it: “It is because the linguistic sign is arbitrary that it knows no other law than that of tradition, and it is because it is founded upon tradition that it can be arbitrary.” (Chandler 2017, p. 31).

Arrow (1974) applied de Saussure’s ideas to organizational contexts, noting that organizational units developed their own specialized communication codes. Zander and Kogut (1995) and Kogut and Zander (1996) subsequently made this argument central to their knowledge-based theory of the firm, whereby the necessity for shared codes at the organizational level justifies the stable relationships that constitute organizations (in contrast to market-based transactions). Members of organizations must share codes to share understanding (Monteverde 1995, Ashforth and Humphrey 1997, Nahapiet and Ghoshal 1998, Loewenstein et al. 2012). Shared codes enable the representation and development of routines (Cohen et al. 1996), organizational knowledge (Garicano and Wu 2012, Loewenstein et al. 2012), collective interpretations (Gavetti and Warglien 2015), and relational contracts (Gibbons and Henderson 2012). Agents that share a code are more likely to implement complementary actions and thus achieve coordination (Argyres 1999). At the same time, it is through repeated attempts at coordination that shared codes develop in organizations (Weber and Camerer 2003, Wernerfelt 2004).

2.2. Prior Literature on Code Differences in Organizations and Why They May Persist

Codes that develop in different organizational units are likely to diverge, as each group adopts its own conventions. This makes communication *across* organizational boundaries challenging (Tushman and Katz 1980). As Weber and Camerer (2003) showed in an influential study, groups that are shuffled after a merger in the laboratory initially face difficulties in harmonizing their codes. This and other experiments show that repeated interaction eventually erodes differences in codes (Fay et al. 2010, Centola and Baronchelli 2015). Observational studies also find that individuals adopt organizational lexicons over time (Srivastava et al. 2017).

However, research on communication and coordination failures also documents instances of significant code differences despite opportunities for extensive interaction (Snook 2002, Jones 2003, Estival and Molesworth

2011). One might suspect this to arise from misaligned incentives: Where there is no mutual interest in communicating, there is no reason for codes to converge. Yet, some of the most vivid descriptions of semantic failures can be found in qualitative analyses of interdepartmental collaboration that occurs within firms (Bechky 2003). Moreover, some accounts of persistent code differences across organizations originate in settings with clear goals and aligned incentives. Snook (2002), for instance, traces the friendly fire that occurred (into the fourth year of joint operations) between the Air Force and the Army to multiple differences in codes between these organizations. Within civil aviation, Jones (2003) documents aircraft crashes that were attributed to differences in codes used by pilots and air traffic controllers. Less immediately lethal but still important are differences in codes in situations involving flight attendants (de Vecchi 2012), architectural design teams (Luck 2013), and medical professionals (Ganiyat et al. 2013). These examples illustrate the long-standing recognition in the organizational communications literature that communication challenges do not disappear with aligned interests. As Axley (1984, p. 431) noted, even if one assumes “that most people are interested in understanding others and in being understood by others as clearly as possible,” nonetheless, “(I)n such an environment, miscommunication and unintentional communication are to be expected, for they are the norm.”

2.3. Types of Differences Across Communication Codes and Their Origins

Bracketing the motivational barriers to code convergence that have been identified in prior literature, we argue that nature of differences in initial codes that agents bring to their communicative interactions create distinctive challenges to convergence. Although, as we noted previously, organizational scholars have long known that communication codes vary across organizations, facilitating communication among insiders at the cost of communication with outsiders, the kinds of differences that can arise when individuals from different organizational units interact have not been scrutinized. We argue that, although codes are arbitrary and specific to organizations, the differences that can arise between them are predictable and are likely to present predictable difficulties for code convergence.

Differences in codes are predictable because codes, like any other convention, solve recurring coordination problems, specifically problems of referring to stimuli in an environment. Clark (1996, 1998) observed that because each community’s lexicon accumulates solutions to coordination problems that arise in that particular community, differences across lexicons

reflect the differences in needs, practices, beliefs, and experiences of social groups. Accordingly, communal lexicons will include labels for stimuli that people outside the community would not recognize, such as, types of mine cavities in the lexicons of miners. Although stimuli in miners' lexicon can uniquely reflect the task environment of miners (in addition to the environment shared with other social groups), the labels in their lexicon may also be unique or shared with outsiders. Thus, the lexicon includes unique stimulus-label pairings (such as "stope," which, for miners, refers to a type of mine cavity), familiar English labels applied to unique stimuli ("raise," meaning a vertical shaft between two levels of a mine), as well as unique labels applied to familiar stimuli ("hoist," in reference to an elevator in a mine).

We build on Clark's work to theorize about the types of differences that can arise between communication codes as a function of their contextual origins. The first level of difference between codes arises with respect to whether codes include references to the same stimuli and whether they use the same set of labels. Organizational units develop codes that allow their members to communicate about the particular stimuli (objects, needs, practices, tools, experiences, etc.) that arise in their specific environment. Furthermore, these codes are likely to be constructed by using labels that are available to the linguistic community within which the organizational unit resides. Thus, codes of communities that exist in similar task environments are likely to have the same stimuli, and codes of communities that use the same vocabulary are likely to have the same labels.

Conversely, when members of organizations from different task environments and different linguistic contexts meet, their codes are likely to be alien to each other. For instance, ground staff and cabin crew typically inhabit different task environments (i.e., ground operations versus caring for passengers in a cabin) and will use different sets of labels that are drawn from vocabularies specific to their occupational communities. However, occasionally they might need to collaborate closely, as in situations involving medical emergencies. They might then discover that one code contains an association like "anhedral" that is alien to speakers of the other code, with neither the label nor the meaning having any counterpart in their prior code. Similar differences in codes, with labels and stimuli that are mutually novel and unfamiliar, are likely to arise in supply chains that entail communication across functions and socio-linguistic groups (Kotha and Srikanth 2013, Srikanth and Puranam 2014).

For codes that share the same set of stimuli and labels, the second level of difference arises with respect

to whether these are associated with each other in the same way. Units that are isolated from each other may well include the same set of labels (because they draw from the same vocabulary of words or abbreviations) and stimuli (because they are in the same task environment) but associate them differently (because communicative interactions are circumscribed by unit boundaries). Continuing with the aviation example, codes of ground staff and cabin crew may include labels like "AP," which means "assistant purser" in one and "aircraft and powerplant mechanic" in the other.

Such jumbled codes with familiar stimuli and labels, but different associations between them, can arise whenever organizational units that operate in similar task environments (industries, functions, occupations) and within the same linguistic context (mostly, but not necessarily, using the same natural language) develop their own codes. Furnas et al. (1987) document the extensiveness of the problem among trained professionals, finding the probability that two typists using the same term actually intend one of the same five generic editorial operations to be 0.41. Jumbled associations are known to arise within the same organization, such as across business units of the same company (e.g., where attrition and "turnover" map differently on to departure of employees, the loss of customers, and sales volume across units; Gulati et al. 2008) and in different branches of the armed forces (e.g., where a key acronym, AOR, for area of responsibility, came to mean the area outside northern Iraq to the Army but the opposite to the Air Force, as they divided their responsibilities during a joint operation; Snook 2002).

In addition to examples of jumbled code with crisscrossed associations of labels and stimuli, there are also examples of codes that differ *only* in set of labels or set of stimuli (Table 1). When codes use different vocabularies in similar task environments, they can contain label clashes, that is, use different labels for the same stimulus. For instance, individuals in different subsidiaries of a multinational organization may use different labels for the same stimuli (Harzing et al. 2011). This can even happen within the same country, as in the example provided by de Vecchi (2012) that we cited above (the labels "comodi" versus "trolley" being used in reference to food carts in two French airlines that merged). Such differences in labeling also arise in academia, as different research groups coin different terms to refer to similar phenomena (Suddaby 2010). The standard paradigm used in psycholinguistic experiments, which uses matching games where individuals describe the same set of stimuli to one another, also leads experimental groups to develop codes that contain label clashes (Weber and Camerer 2003, Koçak and Warglien 2020).

Table 1. Types of Code Differences

	Labels are the same across codes	Labels are different across codes
Stimuli are the same across codes	Jumbled codes For example, turnover, attrition, and churn as used to refer to stimuli like sales, employee loss, or market share loss turnover churn X Y	(Pure) Label clashes For example, comodi vs. trolley used to label food carts in French airline merger comodi trolley X
Stimuli are different across codes	(Pure) Stimulus clashes For example, slide understood to mean different things to assemblers and engineers slide X Y	Alien codes For example, the financial term zombie debt used by credit department and the mathematical term asymptote used in engineering design unit asymptote zombie debt X Y

Note. Boxes represent labels, X and Y represent stimuli, and lines with dashes versus dots represent stimulus-label associations of different agents.

In contrast, different codes can share a vocabulary (set of labels) but use the same labels in reference to different stimuli, thus creating stimulus clashes. This can arise from differences across dialects, as in the example of British and American members of a team that developed mutual distrust because they used and interpreted the word “quite” in different ways (Abrams et al. 2003). Stimulus clashes are likely to arise when functional specialists that speak the same natural language but work with different conceptual schema develop technical jargons, as with architects and civil engineers (Luck 2013), common law lawyers and civil law lawyers (Triebel 2009), or designers and assemblers (Bechky 2003): “To the engineer, the word ‘slide’ connoted the formal term ‘electrode slide’ in the lexicon of the engineering drawing language. For the assembler, on the other hand, the ‘slide’ was one of a class of parts in the concrete language of the machine: parts that physically slide” (Bechky 2003, p. 324). Check-in staff and cabin crew in an airline may both use the label “problem customer,” but these might indicate different properties of the customer. For the check-in crew, this may refer to a person with a complicated booking requiring multiple rounds of verification, whereas for the cabin crew, this could be an unruly passenger.

These distinctions in the ways that communication codes can differ have thus far remained unexamined.

Even when code differences have been identified in similar settings, the types of differences have been unacknowledged. For instance, Suddaby (2010) worries the most about multiple labels referring to the same concepts, warning that it prevents accumulation of scientific knowledge. In contrast, Edwards et al. (2011) describe the most egregious cases of science friction arising from the use of similar terms to refer to different things.

There are indications that these differences merit a closer analysis. For example, miscommunication that arise among pilots who are native speakers of English reverting to colloquial speech rather than sticking to standard aviation English is reported to be harder to fix than training foreign pilots to learn standardized phraseology (Estival and Molesworth 2011). Research on the *lingua franca* in multinational organizations indicates that the different dialects of English (such as formal English taught as a second language and the informal English spoken by native speakers), when spoken by different subunits in an organization, can create more communication breakdowns than different native languages being spoken in the different subunits (Harzing and Feely 2008). A systematic explanation for why some of these forms of code differences should create more persistent communication challenges than others does not yet exist. We attempt to develop such an explanation by

examining the consequences of these qualitative differences in initial codes on the likelihood and speed of code convergence.

2.4. Constraints Common to Communication with Divergent Codes

When individuals with different codes attempt to communicate, failure is inevitable. A miner's reference to a "stope" will not be understood by an outsider to the mining community. A poker player's reference to being "hot" may be misunderstood by a hearer who does not know the poker players' lexicon. Code convergence after such breakdowns in communication can be hindered by constraints that one typically finds in communication contexts, including organizational ones.

One constraint is that individuals need to communicate about stimuli that arise exogenously in their task environment. That is, they cannot typically choose what to communicate about. A second constraint is that codes are mostly private; individuals do not have insight into each other's codes. If they knew how others' codes differed from their own, they could avoid miscommunication by sticking to parts of their codes they know to be shared (Clark 1996, 1998). However, given these two constraints, agents can neither know others' codes *ex ante* nor avoid communicating about stimuli as they arise exogenously.

A third constraint is that it is often difficult, if not impossible, to unambiguously establish joint referents: *what* the different parties are referring to. Absence of joint referents rules out ostensive teaching of words (by pointing to objects and stating their labels) as a possible route to code convergence (Wittgenstein 1953, Quine 1960). In general, a joint referent will be difficult to produce in organizational contexts if agents are in different physical worlds and thus cannot see the stimuli that the other sees or if the stimuli are intangible. MacDuffie (1997) provides an example of the first case, where employees observing paint defects in auto production must communicate the type of defect to others who are not in the same location. In the latter case of intangible referents, even physical collocation cannot help. In an example given by Fine (1995), cooks may sample the same food but may never precisely convey the way it tastes to them. In the illustration of Gibbons and Henderson (2012, p. 1351), managers do not have easily observable things to point to when trying to convey their expectations that employees act "almost as if they were academics" at Merck or become "active problem solvers" at Toyota. The observation of Schein (2006, p. 112) that "[m]embers of a founding group coming together to create a new organization need to learn about each other's

semantic space (even if they start with a common basic language, such as English) in order to determine what they really mean by such abstractions as 'a good product,' of 'high quality,' produced at 'low cost,' to get into the 'market' 'as rapidly as possible.'" underscores the prevalence of the problem of establishing joint referents in organizations.

Despite the inevitability of communication failure when codes diverge, and despite the frequent inability to establish joint referents, see into each other's codes, or control the topic of communication, prior research shows that code convergence can nonetheless arise. The basic process through which this occurs is as follows: communication partners attempt to infer the stimulus referred to by their interacting partner's label, adjust to feedback on whether they have inferred the correct stimulus, and converge on a common code through trial-and-error learning over repeated rounds of interaction. This process has been demonstrated through an extensive body of experimental evidence (e.g., Krauss and Glucksberg 1977, Clark and Wilkes-Gibbs 1986, Garrod and Doherty 1994, Brennan and Clark 1996, Markman and Makin 1998, Weber and Camerer 2003), as well as formalized in computational experiments with simulated agents (e.g., Skyrms 2010, Spike et al. 2017).

We follow these prior studies in assuming aligned interests (in the sense that agents are equally motivated to communicate successfully), exogenously arriving stimuli, private codes, and absence of joint referents. These choices allow us to build on the findings of prior research and focus on our theorized mechanism: the impact of the nature of code differences.

3. Model

3.1. Overview: Code Convergence Through Trial and Error

Formal models of code convergence through learning from interaction are largely built on Lewis's study of conventions (Lewis 1969) and are called "Lewis signaling games" (Skyrms 2004, 2010; Hofbauer and Huttegger 2008; Argiento et al. 2009; Huttegger et al. 2014). Following this literature, we model a language game between two agents, where one agent (playing the role of Sender) labels a stimulus that arises in the environment and the other agent (Receiver) infers the stimulus based on this label. Success entails the stimulus inferred by the Receiver matching the stimulus that the Sender had intended.³ The Sender and Receiver each get a reward r of +1 each if the stimuli match or else zero. They then update their codes based on the reward and swap roles for the next round (i.e., take turns being Sender and Receiver).

3.2. Modeling Initial Differences in Communication Codes

We take a *communication code* to be a mapping between stimuli and the labels used to describe them (Arrow 1974, Kogut and Zander 1996, Argyres 1999, Wernerfelt 2004, Crémer et al. 2007). We follow prior literature in conceptualizing an individual agent k 's code at time t as a two-dimensional matrix (C_{kt}), where $C_{kt}[x, y]$ represents the strength of association between label x and stimulus y (e.g., Nowak et al. 1999, Komarova and Nowak 2001, Smith 2004, Richie et al. 2014). Association strengths lie in $[0,1]$. They connect stimuli (which can be individual entities (objects, events, processes, perceptions), components of such entities, or categories of them), and arbitrary labels (words, phrases, images, or any other symbols) in manner that is private to the agent. Stronger associations will translate into a greater probability (derived from the size of an association relative to others in relevant rows or columns) of a particular label being used to describe a particular stimulus, or a particular stimulus being inferred based on a particular label. We focus on stimulus-label associations and black box the mediating correspondence of stimulus to category/concept and category/concept to label, as prior work on code convergence has also done.⁴

Agents interact in a semantic space that includes all stimuli and all labels that exist in their combined codes. Each stimulus in this space can arise with equal probability (for the Sender to communicate in a message to the Receiver) in any period t . We assume that each agent has one code and all possible stimuli in the task environment are present in each agent's code. We also assume that labels that appear in either agent's C_{k0} are available to both agents, although the associations remain private; that is, each agent knows only its own code (i.e., the pattern of associations) and not that of the other agent. Thus far, our model closely follows prior formalizations of codes in Lewis signaling games (see Spike et al. 2017 for an extensive review) and is consistent with how experimentalists have treated the topic (Brennan and Clark 1996, Weber and Camerer 2003).

The primary novelty of our models, compared with prior studies of code convergence, lies in the different types of ways in which initial codes of agents, developed or learned in different social groups, can vary. We assume that the codes that agents bring to their interaction may or may not include associations for every stimulus and label that exists in the semantic space of their interaction. Figure 1 illustrates the four ideal shapes that the two agents' codes may take at $t = 0$, depending on whether the codes of these agents include associations for the same stimuli (by virtue of the agents having operated in the same task

environments) and whether they have associations for the same labels (by virtue of agents belonging to the same linguistic community). In all instances, Sender and Receiver's C_{kt} have the same dimensions, that is, the same set of labels and stimuli. However, reflecting the individuals' different prior experiences, parts of their codes may have blanks, that is, zero associations, because of not having previously formed mappings for certain stimuli or certain labels.

When agents have *alien codes*, each agent's code has associations between some labels and stimuli that are blank in the other's code, as with stope for someone unfamiliar with mining. The codes we use for the Senders and Receivers in these models have zero overlap in labels and stimuli that have associations.

Jumbled codes arise when both agents have codes with the same labels and stimuli but with different associations between them, as with the example of "attrition" and "turnover". The codes accordingly have no blanks but have different associations, which are set at random, so Senders' and Receivers' C_{k0} have no more than chance overlap in the associations between labels and stimuli.

In order to examine the two types of clashes that arise within jumbled codes, we create two pure cases of label and stimulus clashes. Codes with *only label clashes* (such as "comodi" and "trolley" associated to food carts by different agents) are created with matrices that have twice the number of labels as stimuli, so that each agent has prior associations of each stimuli with a single label and no associations for half the labels. Codes with *only stimulus clashes* (as in the case where different agents refer to different stimuli with the same label slide) are created with matrices that have twice the number of stimuli as labels, whereby each agent uses the same set of labels to refer to a different set of stimuli for which the other agent has blank associations. To portray pure cases of these clashes it is necessary that the code dimensions be imbalanced within agents, because the number of stimuli that rise is always the same for both. We later extend the analysis of label and stimulus clashes to codes with equal dimensions.

3.3. Modeling Interaction

Interaction between agents is modeled as follows: on receiving a stimulus drawn at random at time t , Sender k looks across the association strengths for that stimulus within the set of labels in its C_{kt} and chooses with highest probability the label that has the largest association strength relative to the total sum of associations for that stimulus. This probabilistic selection is based on the *softmax* action selection rule, which is widely used in behaviorally founded models of choice in general (Luce 1959, Sutton and Barto 1998, Posen

Figure 1. Distinct Forms of Initial Code Differences

Alien codes represent situations in which each agent's code contains associations between some labels and stimuli that are blank in the other's code:

1st Agent's Code							2nd Agent's Code						
	S ₁	S ₂	S ₃	...	...	S _s		S ₁	S ₂	S ₃	...	...	S _s
L ₁	1	0	0	0	0	0	L ₁	0	0	0	0	0	0
L ₂	0	1	0	0	0	0	L ₂	0	0	0	0	0	0
L ₃	0	0	1	0	0	0	L ₃	0	0	0	0	0	0
...	0	0	0	0	0	0	...	0	0	0	0	1	0
...	0	0	0	0	0	0	...	0	0	0	0	0	1
L _i	0	0	0	0	0	0	L _i	0	0	0	1	0	0

Jumbled codes represent situations where agents' codes contain associations between the same set of labels and stimuli but the particular stimuli and labels that are matched with one another are different:

1st Agent's Code							2nd Agent's Code						
	S ₁	S ₂	S ₃	...	...	S _s		S ₁	S ₂	S ₃	...	...	S _s
L ₁	1	0	0	0	0	0	L ₁	0	1	0	0	0	0
L ₂	0	1	0	0	0	0	L ₂	1	0	0	0	0	0
L ₃	0	0	1	0	0	0	L ₃	0	0	1	0	0	0
...	0	0	0	1	0	0	...	0	0	0	0	1	0
...	0	0	0	0	1	0	...	0	0	0	0	0	1
L _i	0	0	0	0	0	1	L _i	0	0	0	1	0	0

Codes can also be created to disentangle types of clashes that occur in Jumbled codes - specifically, to contain only **Stimulus clashes** that match the same labels to different stimuli:

1st Agent's Code							2nd Agent's Code						
	S ₁	S ₂	S ₃	...	...	S _s		S ₁	S ₂	S ₃	...	...	S _s
L ₁	1	0	0	0	0	0	L ₁	0	0	0	1	0	0
...	0	1	0	0	0	0	...	0	0	0	0	1	0
L _i	0	0	1	0	0	0	L _i	0	0	0	0	0	1

or only **Label clashes** that use different labels to describe the same stimuli:

1st Agent's Code				2nd Agent's Code			
	S ₁	...	S _s		S ₁	...	S _s
L ₁	1	0	0	L ₁	0	0	0
L ₂	0	1	0	L ₂	0	0	0
L ₃	0	0	1	L ₃	0	0	0
...	0	0	0	...	1	0	0
...	0	0	0	...	0	1	0
L _i	0	0	0	L _i	0	0	1

and Levinthal 2012, Puranam et al. 2015) and signaling models in particular (Franke and Correia 2018). In additional analysis, we find that our conclusions are qualitatively robust to assuming maximization (or greedy action selection). Because the rule is symmetric across agents, we suppress k from the notation, and because the rule does not vary over time, we also

suppress t . The probability that a Sender seeing stimulus n selects label i among the l labels in its C is

$$p(i | n) = \frac{e^{C[i,n]/\tau}}{\sum_{a=1}^l e^{C[a,n]/\tau}}. \quad (1)$$

The parameter τ tunes the imprecision or propensity to make errors: the probability of the agent choosing

an alternative independent of its association strength. Increasing τ makes the agents send labels (or infer stimuli) that are not the ones they should have, given the current state of their communication code (Franke and Correia 2018). We assume τ to be identical for agents in a dyad.

The Receiver in turn uses its own code and the same selection process (softmax) to infer the stimulus given the label i . The probability it will select stimulus m given the label i received from the Sender is

$$p(m | i) = \frac{e^{C[i,m]/\tau}}{\sum_{b=1}^s e^{C[i,b]/\tau}}. \quad (2)$$

We suppressed k and t from the previous notation to keep the exposition simple, but these choice equations operate on agent-specific and time-varying codes at each period in the model. The agents experience successful communication if $m = n$, that is, the stimulus inferred by Receiver is the same as stimulus n observed by Sender.

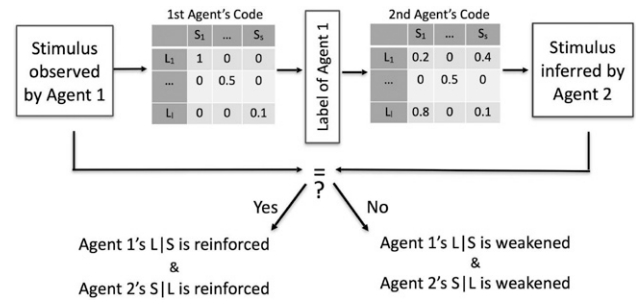
The presence of labels or stimuli with no associations, that is, blanks, in some initial codes (as shown in Figure 1) means that a Sender may need to refer to a stimulus that does not have a prior label or a Receiver may need to guess at a stimulus based on a label that does not have any positive associations. Using Equations (1) and (2) leads Senders seeing unfamiliar stimuli to pick a label at random and Receivers hearing unfamiliar labels to pick a stimulus at random. In additional analysis, we find that identical low values instead of zeros produce the same effects. Alternative decision rules can also be constructed by assuming agents to have different assumptions about the codes and intentions of their communication partners. We also model these in additional analyses.

3.4. Modeling and Measuring Payoffs and Learning

The model represents learning on the job, such that neither agent controls which stimuli arise, successful communication about the stimuli is rewarded, and joint reference to stimulus cannot be established. Formally, the model captures coupled learning (Lounamaa and March 1987, Puranam and Swamy 2016) within a contextual bandit task (Langford and Zhang 2008). (We consider alternative task environments, such as a training/teaching context and mutual visibility of stimuli later.) We display the process of interaction and code adjustment schematically in Figure 2.

Specifically, the task environment produces payoffs to the two agents of $r_t = 1$ each if the stimulus that the Receiver inferred matches the stimulus that the Sender had observed or else zero. The payoffs are a measure of *communication success*, which we track over

Figure 2. Model of Code Convergence Through Interaction



time to observe the rate at which communication becomes more effective (i.e., whether the agents communicate or interact effectively in their task environment). In keeping with the assumption that joint reference to stimuli cannot be established (e.g., because stimuli could be intangible), the Sender never directly observes the stimulus that the Receiver inferred, nor does the Receiver ever see the true stimulus that the Sender tried to convey.⁵ In other words, when there is a match, the agents can deduce that the stimulus inferred and sent were the same, but under mismatch, they receive no information on each other's C_{kt} . For instance, when one employee describes a customer as a problem customer to another, whether the actions the second employee takes in response (a follow-up call or account closure) are aligned with what the first employee intended may be revealed through performance feedback on this action, but the intended referent of a problem customer will remain opaque to the second employee in the event of misalignment. Each agent learns as its C_{kt} is modified after every period, by averaging rewards it has received on selecting a particular cell (Sutton and Barto 1998, Posen and Levinthal 2012).

Psychological studies of learning lend support to imperfect forgetting implied by this model, because they demonstrate that what is learned is influenced by what was previously learned (Shanks 2010), that new memories do not fully overwrite preceding memories (Wixted 2004), and that unlearning (of undesired behaviors) is hard to accomplish fully even with considerable effort (Crossley et al. 2013). Combined with the softmax action selection, which allows for exploration and “endogenously weights the mean belief by uncertainty” (Posen and Levinthal 2012, p. 5–6), this learning process also approximates the normative benchmark established by Gittins and Jones (1979) for a multiarmed bandit problem, because the attraction of a choice depends not only its expected reward (estimated in an unbiased manner through the averaging) but also its uncertainty (i.e., unless the highest association is significantly higher than others in the selected

row or column, there will be exploration). In additional analysis, we also incorporate an update process based on a more general formulation of reinforcement learning (Bush and Mosteller 1955), which allows for tunable rates of learning and also features the behavioral property known as the law of recency, whereby recent feedback is weighed more heavily than past feedback (Erev and Roth 1998, Puranam et al. 2015, Richie et al. 2014).

As agents learn, their codes will become more similar, increasing the likelihood of communication success. In addition to tracking how communication success evolves over interactions, we measure *similarity* of codes over time, as an indicator for the rate at which codes converge. When agents can play either role of Sender or Receiver (and they play each role in alternate rounds in our models), communication success will depend most crucially on similarity between the Sender and Receiver's codes. Similarity is relevant along two dimensions: picking the same label for a given stimulus (labelling stimuli) and finding the same stimulus for a given label (identifying stimuli). We investigate when (and how rapidly) the two agents' codes converge, that is, reach a state of sufficient similarity for them to reliably pick the same stimuli and labels. The duration of the convergence process is the main outcome that we are interested in, as a slower convergence process results in persistent communication difficulties.

We measure similarity between the codes of the agents separately for the two dimensions of the code after every period (i.e., interaction). To calculate similarity in columns across codes (i.e., similarity in labelling given stimulus), we compare whether the strongest associations within columns across the two codes are in the same cells. If they are, then it is more likely that both agents will pick the same label when faced with the same stimulus. We construct an indicator variable that takes on a value of one when the label with the highest association to a given stimulus is identical across agents and zero if not and average this variable across all stimuli. The same approach in transpose can be applied to calculate similarity in rows across codes (i.e., similarity in identifying stimuli given labels). Thus, our measure of similarity for each dimension lies in $[0,1]$. For instance, if the codes of two agents are such that for every stimulus (column), the label with the strongest association is the same for both agents, then their similarity on labeling (picking label given stimulus or Label|Stimulus) is $= 1$. If, however, the stimuli with the strongest association are the same for both agents only for half the labels, then their similarity on identification (Stimulus | Label) is $= 0.5$.

Independent of the dissimilarity of codes *across* agents, miscommunication can arise from lack of crispness *within* agents' codes (Cr  mer et al. 2007).

Lack of crispness here refers to the possibility that within any code, the same label may be associated with more than one stimulus (homonymy) or the same stimuli may have multiple labels associated with it (synonymy), leading to mismatches in communication. To measure crispness in a single agent's code, we measure the difference between the first and next strongest association, averaged across every column (crispness in labeling a stimulus) and every row (crispness in identifying a stimulus). Thus, our measure of crispness for each dimension lies in $[0,1]$. The greater the crispness in labelling, the more likely that the same label is selected by the agent whenever faced with the same stimulus. The greater the crispness in identification, the more likely that the same stimulus is selected by the agent whenever faced with the same label. Within the completely crisp code of an agent, the average difference in associations between largest and second-largest cells will always be $= 1$, whether computed across rows (crispness in identification) or columns (crispness in labeling).

4. Model Implementation and Results

We implemented the model algorithms in Python programming language and computed them typically for $T = 1,000$ periods of interaction. We report results averaged across 10,000 dyads (i.e., runs) for each point in the parameter space we sample. We start with codes that look like those in Figure 1, in which each row and column have at most one cell $= 1$, and all other cells are zero. Initial differences in codes are created through the location of the all zero (blank) and nonzero associations. Our baseline parameter settings for softmax action selection takes $\tau = 0.01$ (high precision in communication). Table 2 summarizes the parameter values for our baseline and supplemental models.

4.1. Jumbled vs. Alien Codes

The chart in the first row of Figure 3 shows communication success over 1,000 periods as agents take turns being Sender and Receiver and adjusting their codes to feedback. Their initial codes are *alien* or *jumbled* (as shown in Figure 1). Overall code dimensions ($s = 10$, $l = 10$) and all model parameters are otherwise identical. They also have identical (and very low, no more than chance) initial similarity.

We find that *jumbled codes* produce strikingly more persistent communication challenges than *alien codes*. By construction, the model eliminates or holds constant factors such as rewards from successful communication, willingness to adjust codes, or the inability to jointly observe the stimulus. Codes are private, ruling out any effects arising from agents compensating for differences (which in any case would operate through adjustment rates, which are constant across cases).

Table 2. Parameter Values

Parameters	Values in baseline models (Figure 3)	Other values used in supplemental models
Code dimensions: $l \times s$ (l , number of labels; s , number of stimuli)	Alien: 10×10 Jumbled: 10×10 Code with only stimulus clashes: 10×20 Code with label only clashes: 20×10	Alien and jumbled: 20×20 (Figure 2.4 in the online appendix) Codes with only label clashes: 40×20 (Figures 2.1 and 2.4 in the online appendix) Codes with only stimulus clashes: 20×20 (Figure 2.2 in the online appendix), 10×10 (Figure 2.3 in the online appendix), and 20×40 (Figure 2.4 in the online appendix) Low precision: 0.1 in the online appendix
Softmax temperature (τ , propensity to make errors)	High precision: 0.01	
Learning rate	$\phi = 1/(N_{x,y} + 1)$, where $N_{x,y}$ is the number of times the specific cell in the code has been selected (and updated) before	$\phi = \{0, 0.3, 0.5, 0.7\}$ in Figure 4
Number of periods: T	1,000	10,000 in Figure 2.3 in the online appendix

Yet, the results indicate that interactions between individuals whose codes contain mutually unfamiliar labels and stimuli are less problematic than those between individuals whose codes contain the same labels and stimuli but with different associations between them. It is as if communication breakdowns between two managers in the same functional areas from the United States and the United Kingdom were somehow harder to overcome than that between French and English managers in a buyer-supplier relationship. We eliminated other possible sources of difference between these cases such as motivation, the ability to establish joint referents, or leveraging other similar portions of code (because we start with codes that are just as dissimilar, as shown in the middle panel in Figure 3). The only possible explanation (because everything else is held constant) can be the nature of initial code differences. The *type*, rather than the degree, of difference between codes must be the culprit.

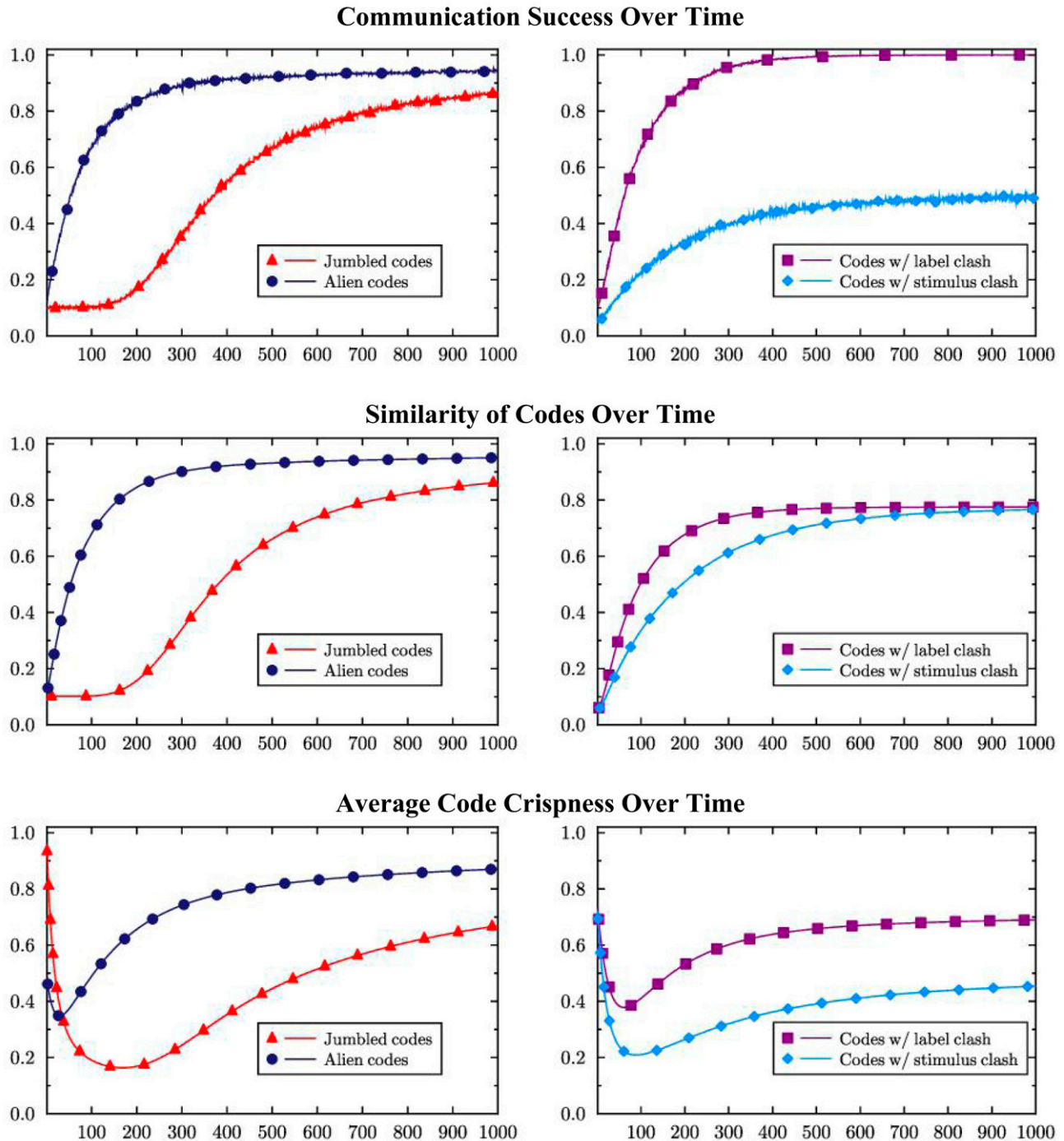
To understand the reasons for this result, it is instructive to consider the crispness of codes. The bottom panel in Figure 3 shows crispness of evolving codes, averaged across agents and across rows and columns over time. In both conditions (*alien* versus *jumbled* codes), similarity of codes across agents improves over time. Initial crispness cannot be identical across jumbled and alien codes because there are blank portions in the latter. However, across both alien and jumbled codes, crispness shows a common pattern of an initial decline, followed by eventual increase. The dynamic relationships between communication success, similarity, and crispness explain this pattern. Initially, because agents start with dissimilar codes, mismatches will be more common than matches. This corresponds to early periods of errors and miscommunications. They produce a period of “unlearning,” as initially crisp but dissimilar codes

become fuzzy (and therefore increase the chance of communication success). As codes continue to become fuzzy, random matches are produced. Reinforcement of these matches leads to increases in both similarity and crispness. This produces the characteristic U-shape of the crispness graphs in Figure 3.

However, this unlearning process is not the same in jumbled and alien codes. Jumbled codes contain clashes: There are nonzero associations to different labels for the same stimuli and vice versa across codes. As Figure 1 shows, for any stimulus, there is an association to a different label in the other code, and for any label, the associations are to different stimuli across codes. For codes to converge, these associations must be unlearned. A period of unlearning (de-crisping) is necessary before random matches can begin to be reinforced to increase similarity and crispness. If two agents are to agree that “turnover” refers to revenue, the one that associates it with departing employee numbers needs to unlearn that association. In contrast, for alien codes, the receiver always draws from the blank portion of their code (Figure 1). A blank association implies no clash and skips the unlearning process. (A nonzero low value instead of zeroes for blank codes creates the same effect.) An agent who has never heard the word attrition before and does not have any labels for the concept of research subjects leaving a study can learn to associate the two without any prior unlearning. As Figure 3 shows, agents starting out with alien codes experience a shorter unlearning period and start crisping up their codes much faster than in the jumbled code case. In the latter, crispness declines much further before it starts to rise compared with the case of alien codes.

We further test this mechanism concerning the need for unlearning in a comparison of jumbled codes

Figure 3. (Color online) Alien Codes Outperform Jumbled Codes; Codes with Pure Label Clashes Outperform Codes with Pure Stimulus Clashes



Notes. Column 1: Alien codes have dimensions 10×10 . Column 2: Codes with pure label clashes have dimensions 20×10 and codes with pure stimulus clashes have dimensions 10×20 .

(where codes clash) to a model where one agent starts with an entirely blank code and a second model where both agents start with blank codes (Figure 1.1 in the online appendix). The latter two both perform better than jumbled codes: It is easier to converge to a common code when one or both agents start out with

a “tabula rasa” than when both agents start out with prior associations.⁶

Given the role that crispness plays in convergence, and in particular its increase as agents unlearn, we perform a further test to check whether the difference between alien and jumbled codes is caused by

differences in aggregate levels of crispness or the specific pattern of differences that arise from clashes (which also produce differences in crispness). Figure 1.2 in the online appendix shows that jumbled codes that are as fuzzy on average as alien codes at the start (the latter appearing fuzzy because of blank patches in the code, the former because of associations less than one) exhibit a dynamic pattern of convergence that looks similar to alien codes but remains at much lower levels, like jumbled codes. Levels of communication success, similarity, and crispness end up much lower than with alien codes and lower than crisp jumbled codes. This confirms that it is the crispness of clashes that slows down unlearning in jumbled codes and not aggregate crispness.

4.2. Disentangling Code Clashes: Stimulus vs. Label Clashes

Do different types of clashes create the same degree of challenge for unlearning? Jumbled codes are likely to have both stimulus and label clashes. We next examine their differential effects. The second column of graphs in Figure 3 show that stimulus clashes are significantly harder to recover from than label clashes, when agents are trying to communicate about exogenously generated stimuli. Again, we can understand this difference in terms of unlearning. In label clashes, only one agent unlearns. For instance, the receiver simply does not know what stimulus is being indicated when the sender says *comodi* in the example from de Vecchi (2012) and guesses at random. Communication failure lowers the crispness of Sender's code significantly (the Receiver's associations are all zero already for the received label). This is a useful failure in the sense that Sender's previous associations are being unlearned, preparing the code for learning new associations. Eventually, through chance, some successful communication occurs, which then increases similarity in the codes.

In contrast, in the case of stimulus clashes, the stimulus that the Receiver infers will not be the one the Sender intended, because the receiver has nonzero associations that clash with those in Sender's code (Figure 1). Therefore, unlearning occurs on both sides. Hearing "slide" in the example documented by Bechky (2003), the receiver *will* reach for the wrong object *and* both agents will reduce their prior associations for slide. There is more unlearning to be done with stimulus clashes compared with label clashes before codes can start to converge. This is evident in the crispness charts, where crispness in the stimulus clash case declines much further before it starts to rise.

Although label and stimulus clashes can exist in codes that contain the same number of labels and stimuli (e.g., jumbled codes), to separate them into pure cases, we modeled the initial codes with greater or

fewer labels than stimuli. However, this way of modeling stimulus and label clashes, with extra labels with no associated stimuli in label clashes and with extra stimuli with no associated labels in stimulus clashes, create additional differences that are separate from the challenge of unlearning. Specifically, we want to partial out two other effects. First, in models of label clashes, with fewer stimuli than in the other types of codes, one may suspect the simpler task environment makes convergence easier. In the online appendix, we show that even when we increase the number of stimuli in the label clash situation to the same number as in stimulus clash, label clashes are easier to resolve than stimulus clashes (Figure 2.1 in the online appendix).

Second, in models of stimulus clashes, both agents' codes include more stimuli than labels ($s > l$). This puts a natural constraint on the relearning process and attainable crispness: with fewer labels than stimuli, these codes have to be fuzzier. In Figure 2.2 in the online appendix, we add blank labels to starting codes so that $s = l$. Empirical studies have documented instances of label creation in order to create a crisper lexicon. For example, when quality analysts at General Motors (GM) plants wanted to develop a common language across departments to help solve paint quality-related problems, they put together a form that included diagrams of various defects and corresponding labels such as "boils, craters, bulls-eyes, sags, runs, orange peel, dings, mars, scratches, cracks, grind marks, powder bumps" (MacDuffie 1997, p. 485). The extended vocabulary likely made it easier to disambiguate types of paint defects. We find that increasing the number of available labels improves communication success and code convergence in situations of stimulus clashes, but the disadvantage relative to label clashes persists for a considerable period.

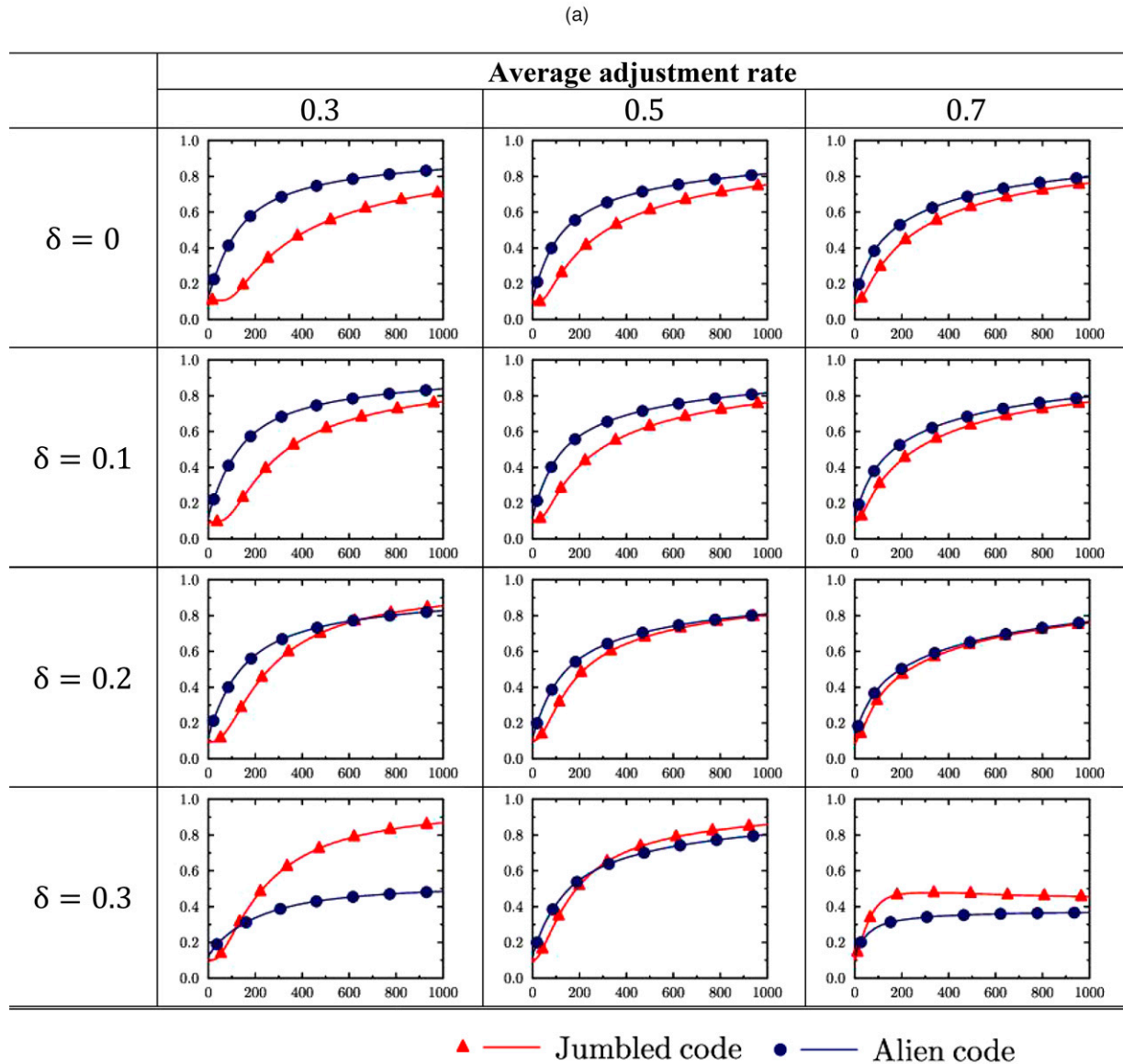
In summary, (a) code differences that involve clashes are harder to overcome than those that do not (even if the codes are just as dissimilar) because clashes need to be unlearned before codes can converge and communication become effective; and (b) among code clashes, stimulus clashes are harder to resolve than label clashes because both agents bear the burden of unlearning in the former.

5. Additional Analysis and Robustness Checks

5.1. Asymmetric Adjustment

Our analysis thus far has assumed that the interacting agents adapt to feedback at the same rate. However, in many social situations, asymmetric learning is likely because of differences in power, status, and authority or structured learning situations with trainer and trainee roles (Haspeslagh and Jemison 1991, Chatman

Figure 4. (Color online) Communication Success over Time with (a) Alien and Jumbled Codes when Agents Adjust Their Codes at Different Rates and (b) Codes That Include Only Stimulus Clashes or Only Label Clashes, when Agents Adjust Their Codes at Different Rates (Reinforcement Models with Tunable Learning Rates (Equation 3))

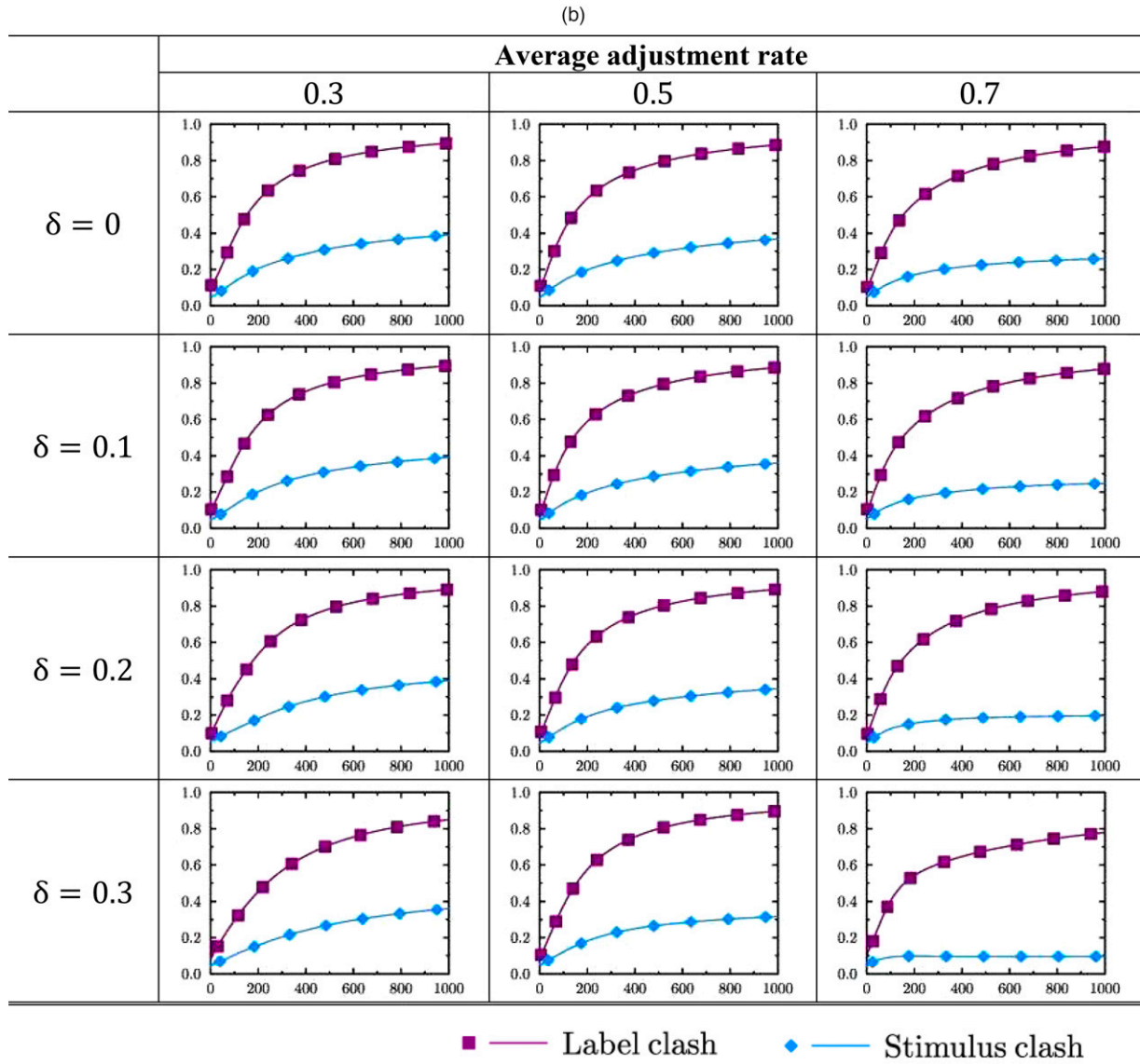


and O'Reilly 2016). Such asymmetries may be adaptive for code convergence and communication. In an experiment on code convergence, Selten and Warglien (2007) found that dyads that endogenously developed differentiated roles so that one individual would adapt to the other were quicker to converge on a shared code. Relatedly, a well-established finding in studies of coupled learning is that the stability of one partner improves aggregate learning (Lave and March 1975, Puranam and Swamy 2016) by avoiding uncoordinated simultaneous adjustments. Because clashes, in particular stimulus clashes, entail both the Sender and Receiver unlearning, we should expect

that asymmetric adjustment improves communication success and convergence in such forms of code difference.

We test this conjecture in models where agents update their codes at different rates. Because the updating based on averaging does not allow for nuance in relative rates of updating, we use a standard version of reinforcement learning with tunable learning rates (Bush and Mosteller 1955, March 1991, Sutton and Barto 1998, Domjan 2014). These models allow us to parameterize learning rates and examine the robustness of our findings to different levels of symmetric and asymmetric learning rates. Specifically, we assume

Figure 4. (Continued)



Notes. (a) Communication success over time with alien and jumbled codes when agents adjust their codes at different rates (reinforcement models with tunable learning rates; Equation (3)). (b) Communication success over time with codes that include only Stimulus Clashes (10×20) or only Label Clashes (20×10) when agents adjust their codes at different rates (reinforcement models with tunable learning rates; Equation (3)).

that the associative strength of the choice n made by the focal agent k in previous period $t - 1$ is updated based on feedback r_t as follows:

$$c_{k,t}[x, y] = c_{k,t-1}[x, y] + \phi_k(r_t - c_{k,t-1}[x, y]), \quad (3)$$

where the parameter ϕ_k , which lies in $[0, 1]$, is the adjustment rate. It captures the willingness or ability of an agent to update the communication code based on most recent feedback. With $\phi_k > 0$, the strength of association between stimulus and label used by an agent is enhanced if the interaction was successful ($r_t = 1$) or else diminished ($r_t = 0$). The averaging rule used

in the baseline analysis is equivalent to setting $\phi = 1/(N_{x,y} + 1)$, where $N_{x,y}$ is the number of times the specific cell in the code has been selected (and updated) before. The associations of the choices not taken remain unchanged in absolute terms though they change in relative terms. To capture differences in the willingness or ability of agents to adjust to feedback, we allow agents in a dyad to have different ϕ_k .

In Figure 4, we examine the robustness of these results to asymmetric rates of learning within dyads, keeping average learning rates constant. Asymmetry in learning rates improves similarity by preventing uncoordinated simultaneous adjustment, that is, by

allowing one agent to (un)learn while the other stays relatively stable. However, asymmetry in learning rates also limits the minimum crispness of codes in the dyad to that of the slower learning agent. The first effect is useful for jumbled codes but not alien codes, whereas the second effect harms alien codes but not jumbled codes, because alien codes have no clashes and lower crispness. Jumbled codes (with clashes) benefit from asymmetric adjustment, whereas alien codes (without clashes) do not; however, even with asymmetry, alien codes continue to enjoy advantages over jumbled codes in terms of rapid improvement in communication success, as do label clashes over stimulus clashes (Figure 4(a)). It takes extreme forms of asymmetry (when at least one of the agents has either a zero learning rate or perfect forgetting—neither of which is plausible) and considerable time for jumbled codes to outperform alien codes (Figure 4(a), bottom row). This suggests that creating asymmetries in learning rates can be a useful intervention for jumbled codes but not for alien codes (we discuss this later), although this may not completely overcome the disadvantage of the former with respect to the latter. The disadvantage of stimulus clashes with respect to label clashes is also robust to asymmetry in learning (Figure 4(b)). In stimulus clashes, the low crispness in labeling (i.e., along the columns of the code) of the slower learning agent acts as a severe constraint on code convergence and improvement in communication.

Another form of asymmetric adjustment is to have the Sender and Receiver in each interaction learn at different rates, regardless of which agent is playing these roles. In additional analysis (results available with authors) using such models, the qualitative pattern is the same as those reported previously. Whether it is the receiver or the sender that learns more slowly relative to their partner, we find that agents that start out with codes containing stimulus clashes have persistent difficulties, whereas codes with label clashes eventually attain high rates of communication success. Codes with label clashes fail to converge to any significant degree only when one of the agents does not learn at all.

5.2. Allowing for Joint Referents

Thus far, we assumed that joint referents are infeasible. Next, we consider what would happen if organizations could somehow reveal the stimulus that the Sender had seen to the Receiver after failures in communication. In the case of the slide, Bechky (2003) observed design engineers visiting assemblers' workspace to sort out the confusion. In general, she noted that tangible definitions will help resolve miscommunication. MacDuffie (1997), in his comparative study of automakers, observed companies invest in production of joint referents to different degrees, from GM's

illustrated glossary of paint defects we described previously to Honda's gathering of staff in the presence of defective vehicles.

We extend the learning models with tunable learning rates we previously introduced to model asymmetry (Equation (3)) to model a version where the true stimulus is revealed to the Receiver after a failed communication, allowing it to imitate the Sender's code. In effect, the Sender's code is no longer private with respect to that particular stimulus. This enables the Receiver to strengthen the association between the true stimulus and the Sender's label and weaken the association between the Sender's label and the stimulus she had inferred:

If $r_{t=0}$,

$$c_{k,t}[x', y] = c_{k,t-1}[x', y] + \phi L_k(1 - c_{k,t-1}[x', y]), \quad (4)$$

where ϕL_k in the interval $[0, 1]$ represents the rate of imitative learning of the association for x' , the label used by the Sender.

In Figure 5, we use reinforcement and imitative learning (Equations (3) and (4)), setting $\phi L_k = \phi_k = 0.3$. We see that convergence with all types of starting codes improve, especially at the start, when the Receiver can copy the Sender's code upon mismatch. However, the long-term improvement in the case of stimulus clashes is negligible.

What would happen if organizations could create joint referents all the time? What happens if organizations take the creation of jointly visible stimuli to the extreme, changing the nature of communication from being on the job (such that stimuli are exogenously generated in the task environment) to being one of demonstration (where a Sender chooses a label and indicates the associated stimulus visibly to the Receiver)? If the Sender reveals both the label and the stimulus to the Receiver, the latter can copy the Sender's code, as in our previous intervention.

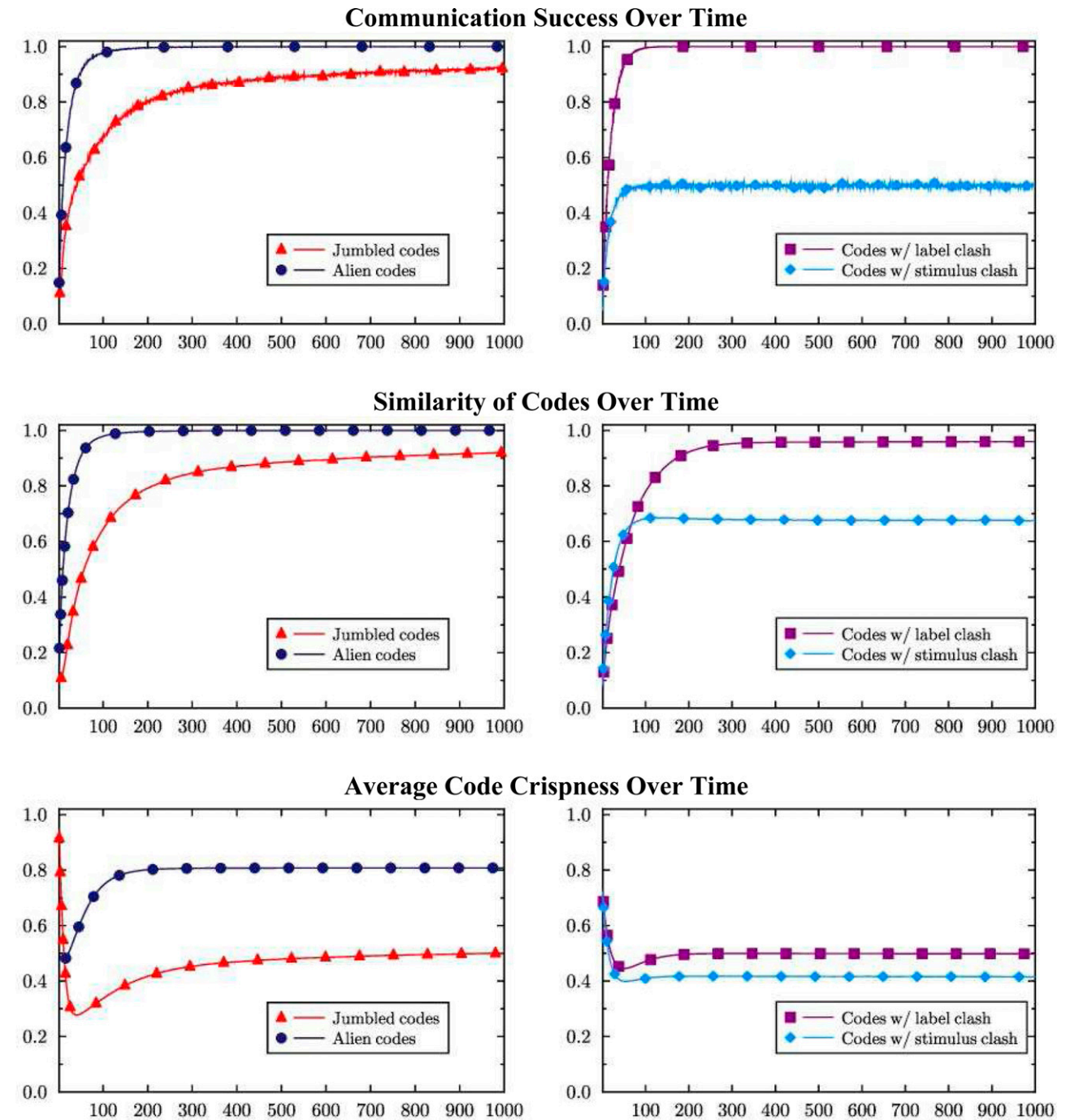
If the Sender only shows the stimulus and the Receiver tries to guess the correct label, as might happen in a context of training by demonstration, the interaction can be modeled as the exact transpose of our baseline model. Applying our results with the appropriate transformation, we can tell that stimulus clashes will be much easier to resolve relative to label clashes in these cases, although jumbled codes will again converge more slowly than alien codes (because the transpose would not affect the relative location of the blanks in the codes). This points to a third possible intervention that organizations facing stimulus clashes can use, namely, setting up demonstrations. This, of course, is not a trivial challenge.

It is in general far easier to transmit labels (words, descriptions, images) between actors than stimuli, which may be intangible (e.g., a matter of taste, a

judgement of trustworthiness or risk) or impossible to present at will for other reasons. Furthermore, actors can create new labels (e.g., new words/phrases/icons), but the stimuli in the task environment are typically exogenous. On-the-job training programs that spell out associated labels for intangible stimuli that exogenously arise in the task environment are common (e.g., novice police trainees learning the meaning

of “asshole” in Van Maanen (1978) or medical students learning the meaning of “crock” in Becker (1993)), but it is much harder to do the reverse by producing stimuli to illustrate the meaning of a given label. This is why the structure assumed in standard Lewis signaling games such as the one we deployed is one of exogenous, nonmutually visible stimuli and label transfer among actors.

Figure 5. (Color online) Code Convergence with Joint Reference (Tangible Definitions) After Mismatches (Reinforcement Models Given in Equations (3) and (4))



5.3. Alternative Pragmatics

We assumed that when a sender encounters an unfamiliar stimulus (in the alien codes situation), it picks a label at random. This corresponds to an assumption that the Receiver might be using any of the labels in the code for this unfamiliar stimulus. One alternative to this rule (in addition to generation of a new label, which we have already considered), is to pick an unfamiliar label at random, from among those labels that are not currently already associated with a stimulus. This could be motivated by assuming that senders and receivers have some degree of visibility into each other's codes and how they choose labels and infer stimuli, on the basis of which they might make some inferences (e.g., see the principle of contrast in studies of child language development; Clark 1990).⁷

We therefore computed variations of our model with these alternative choice rules (baseline for Sender and contrasting (i.e., avoiding already associated stimuli or labels) for Receiver, vice versa, and contrasting for both Sender and Receiver). In all variations, code convergence with alien codes improved significantly over the baseline, thus widening even further its advantage over jumbled codes. Codes with stimulus clashes improved significantly when Senders generated and used new labels for unfamiliar stimuli, although they still underperformed codes with label clashes. Our primary findings that (a) jumbled codes (and clashes) are harder to resolve than alien codes (and blanks), and (b) stimulus clashes are harder to overcome than label clashes remain unchanged across all these variations in (pragmatic) choice rules.

5.4. Asymptotic Properties

Given enough interactions, codes of the same dimension should attain comparable levels of communication success and similarity because of the exploration parameter in our models ($\tau > 0$). We find that jumbled and alien codes achieve similar levels of code convergence and communication success in the long run, although the effects of initial code differences remain visible well into thousands of rounds of interaction (Figure 2.3 in the online appendix). Even stimulus clashes, once we allow for the same dimensions of code as alien or jumbled codes (with added labels to make $s = l$), eventually perform as well as jumbled and alien codes, although the effects of initial code differences erode only after thousands of periods. Task environments vary in how patient they are in tolerating communication failure. Furthermore, even the motivation to attempt further communication may be endogenous to past success (something we ignore for simplicity). Therefore, these differences that persist across such large periods are likely to be even more significant than what is indicated in our model, because communication (or even the communicators)

may simply cease to exist after an extended period of poor communication.

5.5. Supplemental Models with Different Parameter Values

The qualitative conclusions from our primary analysis (the disadvantage of jumbled codes and in particular stimulus clashes) do not change at higher softmax temperatures (Figures 3.1 and 3.2 in the online appendix). Increasing exploration does not increase eventual crispness, because it also slows down communication success and the growth in crispness that it produces. We also verify that greedy action selection (zero exploration) does not alter the qualitative pattern of results (results available from the authors), and in fact, differences even between codes of the same dimension persist asymptotically. The results also remain qualitatively the same with increases in the size of codes that preserve their proportional dimensions (Figure 2.4 in the online appendix). In additional analysis (results available from authors), we also verify that changing the zeroes in blank codes to nonzero low values effectively introduces an unlearning challenge even into blank codes, but as long as blank codes pose less severe challenge than clashes (i.e., we use some constant nonzero but relatively small values instead of zero), the results are qualitatively unchanged.

6. Discussion

As work becomes more distributed and as diversity in the backgrounds of interacting employees increases, the question of how individuals with different communication codes can converge on a common code gains greater importance. Our model and findings suggest that the difficulty of code convergence depends on the nature of differences in initial codes. When agents' prior codes contain identical stimuli and labels that are mapped in different ways, creating clashes across codes, convergence to a common code and communication success requires that agents first unlearn their prior codes. The need for unlearning when codes clash will create friction in communication success and delays in code convergence even without any misalignment in incentives. In contrast, when the dissimilarity in codes arises from unfamiliar stimuli and labels (and, by implication, contains fewer clashes between familiar stimuli and labels), the challenge of unlearning is modest.

Among the types of clashes, we find that agents who developed their communication code in different task environments by attaching the same labels to different stimuli have particularly persistent difficulty with code convergence. This finding offers a potential explanation for some otherwise puzzling patterns. For instance, it is harder to fix miscommunications that

arise from native English speakers reverting to colloquial speech instead of sticking to standard aviation English than it is to train foreign pilots to learn standardized phraseology (Estival and Molesworth 2011). Different dialects of the same language (such as formal English taught as a second language and the informal English spoken by native speakers), when spoken by different subunits in an organization, can create more communication breakdowns than different native languages being spoken in the different subunits (Harzing and Feely 2008). Our results show that it is important in such contexts to develop a formal process to agree on words and what they mean, even if it seems counterintuitive for native English speakers.

In contrast, individuals who have been socialized in similar task environments, such as an occupation, a profession, or an academic discipline, yet use different vocabularies, are likely to find it easier to develop common codes through interactions. When dissimilarity in codes arises from alien stimuli/labels or contain label clashes but not stimulus clashes, the incentives to communicate successfully are sufficient to stimulate code convergence, because the receivers do not have to unlearn associations for the senders' new labels. These results imply that communication challenges in postmerger integration, buyer-supplier relationships, and research and development alliances may all be *less* severe than in cross-functional collaboration within a firm (which may involve the same set of labels for very different stimuli). This adds significant nuance to the intuitions that collaboration between different functions is often a source of communication challenges in organizations (Dougherty 1992, Bechky 2003) and that the boundaries of the firm are where communication difficulties arise (Arrow 1974, Tushman and Katz 1980, Zander and Kogut 1995). Cross-functional collaboration within a firm may not only be more challenging than collaboration within functions but may also be more challenging than cross- or within-functional collaboration between firms.

These two findings, that jumbled codes lead to greater persistence in miscommunication relative to alien codes and stimulus clashes are harder to resolve than label clashes, are robust to a range of parameter values, asymmetry in learning rates across agents or sender/receiver roles, allowing joint referents, and alternative pragmatics. That said, differences across types of initial codes do disappear over very long time horizons. This assumes, of course, that agents do not give up trying to communicate. Another way in which our results could be inverted is to allow for demonstration; for agents to choose which labels to convey to the other by pointing out stimuli. As we noted, there are a limited set of circumstances in which this is feasible. Finally, an extreme degree of

asymmetry in adjustment rates (if one of the agents either does not learn at all or has perfect forgetting) combined with sufficient time can also reverse our results, but this too does not seem likely under normal conditions.

Outside the artificial simplicity of a model, there are a number of additional reasons stimulus clashes may present greater challenges than other types of code differences. For instance, we assumed that feedback on communication success is immediate. If there is a delay in feedback, individuals who use the same labels for different stimuli may assume that they are in agreement (Gumperz and Tannen 1979). This would delay convergence further. We also assumed that agents can perceive previously unfamiliar stimuli when they arise in a shared task environment. However, perception of some stimuli may be difficult without prior exposure or training. This would also create additional challenges for code convergence in the case of stimulus clashes.

Our results also indicate that interventions that create tangible referents are always useful to overcome code differences while creating glossaries with new labels or asymmetric learning situations are useful for dealing with the clashes inherent in jumbled codes (new labels are unnecessary for alien codes and asymmetry is even harmful). Yet these benefits appear to primarily be through mitigation of label clashes; our model predicts that these interventions are not very effective against stimulus clashes. An implication is that organizations that suspect stimulus clashes to arise (i.e., when communication occurs between groups that operate in different task environments but use the same natural language) may prefer alternatives to code integration. One alternative is to use interfaces such as translators who are aware of code differences and are able to switch between multiple codes (Tushman and Katz 1980). Another is to reduce the need for communication by modularizing work (Sanchez and Mahoney 1996, Baldwin and Clark 2000). Yet another is to change the nature of communicative interaction such that agents demonstrate labels by exchanging stimuli rather than exchange labels to communicate about stimuli.

Our study also contributes to a rapidly expanding field of research on language in organizations. Early studies that marked distinct cognitive associations by use of distinct word use (e.g., Dunn and Jones 2010) have recently been significantly extended using text analytic approaches. Comparing stimulus-label associations in different contexts and time periods, these have demonstrated that differences in codes can arise from differences in associations rather than just vocabularies (Piezunka and Dahlander 2015, Srivastava et al. 2017). Recent experimental research on code convergence has also pointed to different degrees of divergence

(Koçak and Warglien 2020). The findings of the current study, while building on prior observational and experimental research, remain theoretical conjectures. We look forward, in particular, to the use of measures of crispness and similarity of codes to verify or refute the key predictions of our model in further studies.

How code differences are bridged by learning through interaction is a particular form of organizational learning (Argote 2013, Argote et al. 2020). Our results also contribute to this literature by highlighting the importance of unlearning- and the different challenges posed to unlearning by qualitatively different initial conditions. Although our analysis has been focused on unlearning in communication codes, we believe the spirit of our results may also be relevant to any form of multiactor learning in a context that involved the formation of conventions. Consistent with the structuralist approach that sees the emergent conventions of language as a general model for culture (Chandler 2017), Weber and Camerer (2003) use convergence of communication codes as a general model of cultural codes. Cronin and Weingart (2007) demonstrate that representational gaps—inconsistencies between individuals' definitions of a team's problem—limit knowledge integration and increase likelihood of conflict. These gaps can concern the end goals, main constraints, problem elements, or operating rules of a given problem. One can imagine coordination difficulties arising from either unfamiliarity of key concepts or clashes in the way different concepts are mapped to one another in causal or correlational chains. Our models predict that knowledge integration would be hardest when codes clash rather than being merely different, particularly when translated terms carry different meanings across cultures (Pratt 1991). The distinction we make between the types of differences in codes may be applied to study conceptual differences across cultures, such as between different occupational communities or the thought worlds of scientific and indigenous communities (Fleck 2012, Ludwig 2016). Further research in this area can help understand constraints on knowledge integration and point to potential interventions that can be used to facilitate it.

A promising line of empirical research is to consider how social identities and communication codes can jointly facilitate coordination in and across organizations (Kogut and Zander 1996). To the extent that identities are contextually anchored to particular task environments and linguistic communities, differences in identity should be correlated with differences in communication codes. This has at least two consequences in terms of signaling and in terms of motivation. First, the association of identities with codes means that use of a particular code can convey membership in a collective. Just as languages, dialects, and

speech patterns function as salient markers of ethnic and national groups (Bourhis and Giles 1977), differences in communication codes, and in particular, vocabularies, can mark differences in cultural origins and background knowledge of communication partners. Conversely, other markers of identity (for instance different uniforms used across organizational units) can make it easier to notice differences in codes and prompt people to seek translators or initiate disambiguation right away, without waiting for miscoordination to arise. This would be especially useful when codes contain stimulus clashes, which would not be apparent until after they cause miscommunication.

Although our analysis focuses on cases where codes are private and agents do not have awareness of code differences, relaxing this assumption would permit more complex models of audience-specific communication. For instance, individuals who are aware of stimulus clashes can use them strategically to communicate different messages to different subgroups within an audience (Eisenberg 1984). Or, individuals can engage in code switching in their interaction (Nilep 2006, Hall and Nilep 2015), exploiting label clashes to signal sameness or difference in identity (Bucholtz and Hall 2004, 2005). Code switching becomes possible when there is a shared meta-code about how codes are associated with particular socio-linguistic groups or collective identities. We can therefore expect it to be used successfully as a communicative device in societies with stable social groups whose distinct codes are easily identifiable to outsiders.

Second, identity differences may also play a role in shaping the motivation to unlearn existing codes and adopt new ones, thereby influencing the trajectory of code convergence. Individuals may be more willing to learn and use the code of groups they want to belong. Auer (1999) argued that two languages merged to form a new one when, over time, bilingual groups that code-switch and mix the two languages decide to mark identity boundaries with a new language of their own or orient specifically to in-group members. The transition to a new, different, common code happened only when a common identity motivated it. In organizational settings, cross-unit codes may fail to form if people are psychologically attached to their subunits and specialized codes (Kumar and Nti 1998). Our models suggest that identification with the larger organization may be a necessary precursor to establishment of a shared code especially when prior codes include clashes, which make convergence particularly difficult.

Clearly, much remains to be learned about the co-evolution of communication codes and related phenomena such as culture and identity. We believe our analysis provides a useful set of tools to conceptualize

these and qualitatively distinguish variants among them, as well as a platform for studying the dynamics of how differences across actors in terms of these constructs are bridged to enable collective action.

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Endnotes

¹ The terms label and stimulus roughly correspond to signifier and signified used by de Saussure (1998). Stimuli may be single entities (e.g., objects), tangible or intangible (e.g., perceptions), and features of entities. Labels need not be single words; they can be phrases, descriptions, or images.

² We think of stimuli being mapped to labels through concepts. The same labels may be understood differently by different receivers because the mappings are different or because the concepts themselves are different (my red need not be the same as your red). We do not consider the latter case of conceptual divergence in this paper.

³ A more elaborate description may involve the receiver taking an action based on the inferred stimuli, which if appropriate given, the stimuli actually perceived by the sender results in a benefit to both sender and receiver. We do not formally model the step between inferring the stimulus and taking an action by the receiver.

⁴ Formally, we can think of a communication code as the multiplication of two matrices: stimulus-categories and categories-labels, which yields a code as a stimulus-label matrix. Code differences can thus result in differences across individuals in either of the two matrices being multiplied; conversely, it is also possible to obtain identical codes even if individuals differ in both. Thus, coordinated action may fail despite shared communication codes (i.e., because of differences in the underlying concepts that link the same stimuli to the same labels across codes; Murphy 2002). However, shared communication codes are important enough for coordinated action to merit careful analysis of the conditions under which they arise (or not) (March and Simon 1958, Arrow 1974).

⁵ When a joint referent is available, code convergence can also happen through imitation (also called naming games; Steels 1998, 2015; Baronchelli et al 2006; de Vylder and Tuyls 2006; Centola and Baronchelli 2015; Tria et al 2015). In our analysis, we later introduce the possibility that organizations make a joint referent available after failed communication (Skeyrms 2010, Spike et al 2017).

⁶ The case where one agent starts with a blank code has an advantage over both agents starting out with blank codes because preexisting associations in one code can serve as attractors for the other.

⁷ For instance, the Sender might think that the Receiver uses labels that are unfamiliar to the Sender for stimuli that are unfamiliar to the Sender. Similarly, on the reception side, when the Receiver gets an unfamiliar label, it may infer an unfamiliar stimulus at random, from among those stimuli that are not currently already associated with a label. We might expect this behavior when the Receiver knows that the Sender knows their code and assumes that the Sender would have used a label that the Receiver associates with the stimulus if there had been such a label.

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